

5G Network Quality-of-Service (QoS) Analysis Dashboard

Academic Project for Data Analysis & Visualization and Big Data Fundamentals

Project Team

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1. Project Overview

The project focuses on analyzing **5G mobile network Quality of Service (QoS)** using network performance data and presenting the resulting insights through an interactive analytics dashboard.

The project follows an end-to-end data analytics workflow:

Network QoS Dataset → Data Preprocessing → Statistical Analysis → Exploratory Data Analysis → QoS Performance Analysis → Machine Learning Prediction → Big Data Processing → Interactive Dashboard

The main QoS parameters considered include:

Download Speed, Upload Speed, Latency, Jitter, Signal Strength and Network Technology.

The overall aim is to understand network performance, identify relationships between QoS parameters, analyze differences across carriers and network technologies, and eventually develop predictive and visualization-based tools for network performance monitoring.

2. Project Objectives

The project objectives established during Review 1 are:

- Analyze historical mobile network QoS data.
- Identify factors affecting network performance.
- Identify relationships among important QoS metrics.
- Perform statistical analysis and visualization of network behavior.

- Develop predictive models for selected QoS metrics.
 - Build an interactive analytics dashboard for network monitoring.
 - Demonstrate Big Data processing using Apache PySpark.
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3. Review 1 — Problem Definition and Project Planning

3.1 Problem Identification

The increasing deployment of **4G and 5G networks** has resulted in rapid growth in mobile data traffic and connected devices. Network performance can vary because of factors such as **network congestion, user mobility, signal conditions and geographic location**.

Maintaining consistent **Quality of Service (QoS)** is therefore important for both users and telecom operators.

For example, a user may experience slow video loading even when connected to a 5G network. This can be related to factors such as throughput, latency, signal strength or network conditions.

Therefore, the project was defined around the analysis of network measurements to understand and eventually predict QoS behavior.

3.2 Key QoS Metrics Identified

The major QoS parameters identified in Review 1 were:

Download Speed – Rate at which data is received by the user.

Upload Speed – Rate at which data is transmitted from the user.

Latency – Time taken for data to travel between endpoints.

Packet Loss – Percentage of packets that fail to reach their destination.

Jitter – Variation in packet delay.

Signal Strength – Strength of the received wireless signal.

Network Technology – Technology used by the connection, such as 3G, 4G or 5G.

3.3 Dataset Selection

The project selected mobile network QoS data containing parameters such as download speed, upload speed, latency, signal strength, jitter, carrier and regional/network information.

The datasets were selected because they provide the network-performance parameters required for:

- QoS analysis
- Statistical analysis
- Exploratory Data Analysis
- Machine Learning
- Dashboard development

Review 1 Status

Completed:

Problem definition → Objectives → QoS metrics → Dataset identification → Overall project workflow

Review 1 established the **scope and technical direction** of the project.

4. Review 2 — Dataset Understanding and Statistical Preprocessing

Review 2 moved the project from the planning stage into **actual dataset analysis and preprocessing**.

The focus was:

Understand Data → Check Quality → Prepare Data → Analyze Statistics → Explore Relationships

The review covered dataset inspection, data quality, missing-value treatment, outlier analysis, feature scaling and statistical distribution analysis.

4.1 Dataset Inspection

The dataset was inspected using:

head() → **tail()** → **info()** → **describe()**

These operations were used to understand the dataset structure, feature types, completeness and basic statistical characteristics.

The dataset contains **50,000 network measurement records and 21 features**.

The inspection established the initial structure required for further preprocessing and analysis.

Status

Completed

5. Data Quality Analysis

5.1 Outlier Detection

Outliers were analyzed using **Tukey's IQR method**.

The method calculates:

$$\text{IQR} = Q3 - Q1$$

The acceptable range was defined using:

$$\text{Lower Limit} = Q1 - 1.5 \times \text{IQR}$$

$$\text{Upper Limit} = Q3 + 1.5 \times \text{IQR}$$

A total of **12 numerical QoS/features** were analyzed.

The analysis detected:

0 statistical outliers

Therefore, no outlier removal was required based on the IQR criterion at this stage.

Status

Completed

6. Missing Value Treatment

Missing-value handling was studied using two primary approaches:

Mean Imputation and **Median Imputation**.

Mean represents the average value:

$$\text{Mean} = \Sigma x / n$$

Median represents the middle value after sorting the observations.

The implementation used:

`mean()` → `median()` → `fillna()`

The distribution's **skewness** was also considered when selecting an appropriate method.

Mean can be influenced by extreme values, whereas median is more resistant to extreme observations.

Status

Completed

7. Feature Scaling

Feature scaling was studied to make numerical features comparable when they have significantly different ranges.

Two approaches were considered:

Min-Max Normalization

$$x' = (x - x_{\min}) / (x_{\max} - x_{\min})$$

This maps the feature to a fixed range, generally **0 to 1**.

Standardization

$$z = (x - \mu) / \sigma$$

This transforms the feature relative to its mean and standard deviation.

The resulting distributions were compared to understand the effect of scaling on the QoS features.

Status

Completed

8. Statistical Distribution Analysis

The statistical behavior of the QoS features was analyzed using:

Mean, Variance, Skewness and Kurtosis.

Mean describes the central or average value.

Variance describes how widely values are spread around the mean.

Skewness describes whether the distribution is relatively balanced or stretched toward one side.

Kurtosis describes the extreme-value/tail behavior of the distribution compared with the normal reference.

The project uses **Fisher's excess kurtosis**, where:

Excess Kurtosis = Pearson Kurtosis – 3

Therefore:

Normal Distribution → Excess Kurtosis = 0

The statistical distribution analysis provided an initial understanding of the shape and spread of the QoS features.

Status

Completed

9. Review 2 Completion Summary

At the end of Review 2, the following modules have been established:

Module	Status
Problem Definition	Completed
Project Objectives	Completed
Dataset Identification	Completed
Dataset Inspection	Completed
Data Type Analysis	Completed
Missing Value Analysis/Treatment	Completed
IQR Outlier Analysis	Completed
Feature Scaling	Completed
Statistical Distribution Analysis	Completed
Relationship/EDA Analysis	Next Stage
QoS Benchmarking	Future
Machine Learning	Future
Big Data Processing	Future
Interactive Dashboard	Future

10. Review 3 — Exploratory Data Analysis and QoS Performance Analysis

Planned Focus

Review 3 will move from individual feature statistics toward **relationships and performance patterns between QoS parameters**.

Module 1 — Exploratory Data Analysis

The next stage will include:

- **Univariate Analysis** — studying individual QoS features.

- **Bivariate Analysis** — studying relationships between two features.
- **Multivariate Analysis** — studying multiple QoS parameters together.
- Histograms
- Box plots
- Scatter plots
- Correlation analysis
- Correlation heatmap

The purpose is to identify meaningful patterns and relationships within the network data.

Module 2 — QoS Performance Benchmarking

The QoS metrics will be compared across different dimensions:

Carrier → Network Technology → Location → Time

The analysis will focus on:

- Download Speed
- Upload Speed
- Latency
- Jitter
- Signal Strength

For example, the project can examine whether QoS performance changes between **4G, 5G NSA and 5G SA**, or whether network performance changes between different carriers or time periods.

Module 3 — Temporal Analysis

Network performance will be analyzed across time to identify patterns such as:

Peak usage periods → Performance variation → Possible congestion periods

This can help connect network measurements with real-world situations such as higher network usage during evening hours.

Review 3 Expected Outcome

Review 3 will establish the **relationships and performance patterns** within the QoS dataset and prepare the relevant features and insights for predictive modeling.

11. Review 4 — Machine Learning, Big Data and Dashboard Integration

Module 1 — Machine Learning Prediction

The next stage will introduce predictive modeling for selected QoS metrics, primarily:

Download Speed and Latency

Potential regression models include:

- Linear Regression
- Decision Tree
- Random Forest
- Gradient Boosting
- XGBoost

The models will be evaluated using:

MAE, RMSE and R^2

MAE

Mean Absolute Error

Measures the average prediction error in the original unit.

Lower MAE → smaller average prediction error

RMSE

Root Mean Square Error

Gives greater importance to larger prediction errors.

Lower RMSE → fewer/lower large prediction errors

R^2

Coefficient of Determination

Measures how much variation in the target is explained by the model.

Higher R^2 → greater explained variance

12. Big Data Processing

The project will demonstrate **Apache PySpark** for distributed processing of network data.

The Big Data module will cover the **5 V's**:

Volume — amount of network measurement data.

Velocity — speed at which network measurements can be generated.

Variety — mixture of numerical, categorical, temporal and Boolean data.

Veracity — reliability and quality of the data.

Value — useful insights obtained from the data.

PySpark DataFrames will be used for operations such as:

Filtering → Transformation → Aggregation → Grouped Analysis

The purpose is to demonstrate how the same type of network analytics can be extended to larger-scale telemetry data.

13. Interactive QoS Dashboard

The final stage will integrate the analytical results into an **interactive web dashboard**.

The planned dashboard modules include:

Overview

Key network KPIs and overall performance summaries.

QoS Analysis

Performance comparison across:

Carrier → Technology → Time → Location

Prediction

Machine learning model performance, predictions and relevant model results.

Big Data

PySpark processing results and the **5 V's of Big Data**.

The dashboard is planned using:

Next.js + React + TypeScript + Recharts

14. Overall Project Roadmap

Review 1 — Foundation

Problem Definition → Objectives → QoS Metrics → Dataset Selection

Status: Completed

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Review 2 — Data Understanding

Dataset Inspection → Data Quality → Missing Values → Outliers → Scaling → Statistical Analysis

Status: Completed

↓

Review 3 — Analytics

EDA → Correlation → QoS Benchmarking → Carrier Analysis → Technology Analysis → Temporal Analysis

Status: Planned / Next Stage

↓

Review 4 — Prediction & Integration

Machine Learning → Model Evaluation → PySpark Big Data Processing → Dashboard Integration
→ Final Insights

Status: Planned

15. Final Expected Outcome

The completed project will provide an end-to-end **5G QoS analytics platform** that moves from raw network measurements to meaningful network insights.

The final workflow will be:

Raw Network Data

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Data Cleaning & Preprocessing

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Statistical & Exploratory Analysis

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QoS Performance Benchmarking

↓

Machine Learning Prediction

↓

Big Data Processing

↓

Interactive Dashboard

The final system is intended to demonstrate how network telemetry can be transformed into **statistical insights, predictive information and interactive visualizations** for understanding mobile network performance.

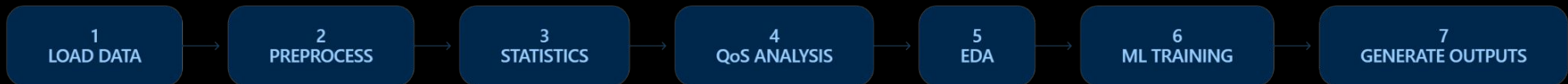
REVIEW 2

Mobile Network Quality-of-Service Analysis and Prediction

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This review focus on:

Understand Data → Check Quality → Prepare Data → Analyze Statistics → Explore Relationships

- **Dataset Understanding** — structure, features and basic statistics
- **Data Quality** — missing values and outlier identification
- **Data Preparation** — missing-value treatment and feature scaling
- **Statistical Analysis** — mean, variance, skewness and kurtosis
- **Exploratory Data Analysis** — univariate, bivariate and multivariate patterns

Outlier analysis

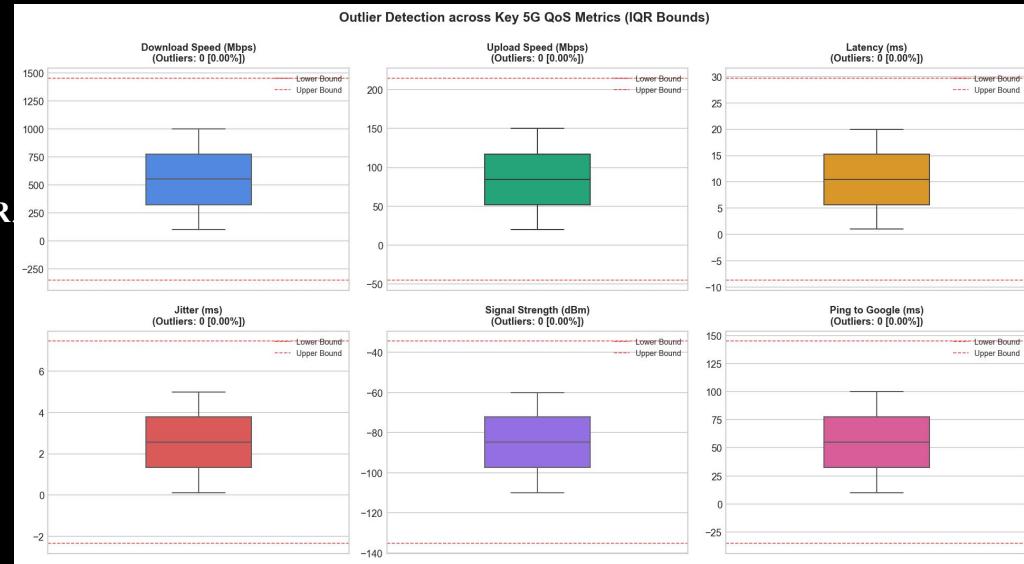
Method

- Used **Tukey's IQR method** to identify statistical outliers.
- For each numerical feature, calculated **Q1, Q3 and IQR**.
- Outlier limits were defined as **$Q1 - 1.5 \times IQR$** and **$Q3 + 1.5 \times IQR$** .

Key Results

- **12 numerical QoS/features analyzed**
- **0 outliers detected across all 12 features**
- **Therefore, 0% of the dataset was classified as outliers for these features.**

IQR analysis found no statistical outliers in the 12 numerical features, indicating that the dataset does not require outlier removal at this stage.



Dataset Inspection - head(), tail(), info(), describe()

Basic inspection established the dataset structure, feature types, completeness and statistical baseline before preprocessing and EDA.

```
inspection_data = {
    'shape': df.shape,
    'head': df.head(),
    'tail': df.tail(),
    'info': info_str,
    'describe': df.describe(),
    'dtypes_df': dtypes_df,
    'unique_df': unique_df,
    'numerical_cols': num_cols,
    'categorical_cols': cat_cols,
    'boolean_cols': bool_cols,
    'temporal_cols': ['Timestamp'] if 'Timestamp' in df.columns else []
}
```

Method / Output	Purpose	Outcome from Dataset
`shape`	Identifies dataset size	50,000 × 21
`head()`	Inspects first records	Confirms initial data structure
`tail()`	Inspects last records	Confirms end-of-dataset records
`info()`	Checks data types & completeness	21 features, no null entries
`describe()`	Provides statistical summary	Mean, Std, Min, Q1, Median, Q3, Max
`dtypes_df`	Categorizes column data types	Numerical, categorical, Boolean & temporal
`unique_df`	Checks unique values	Identifies categories/cardinality

Missing Value Treatment: Mean vs Median

```
df_mean_imputed = df.copy()
df_median_imputed = df.copy()

num_cols = df.select_dtypes(include=[np.number]).columns.tolist()
imputation_records = []
```

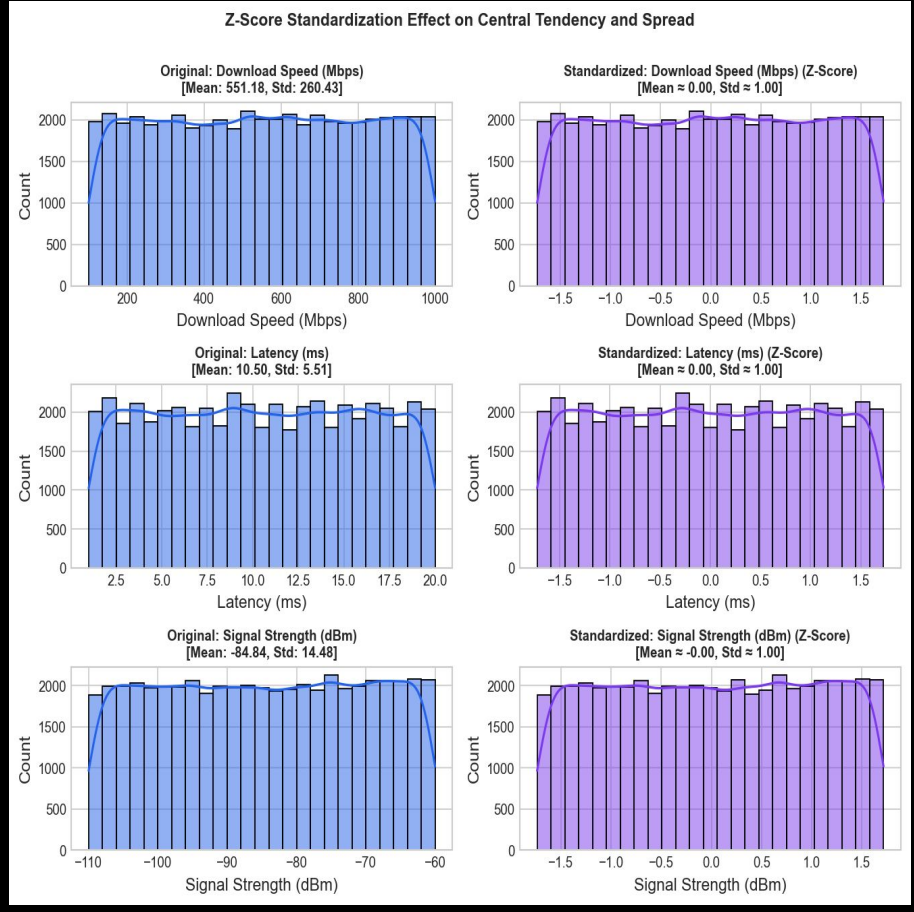
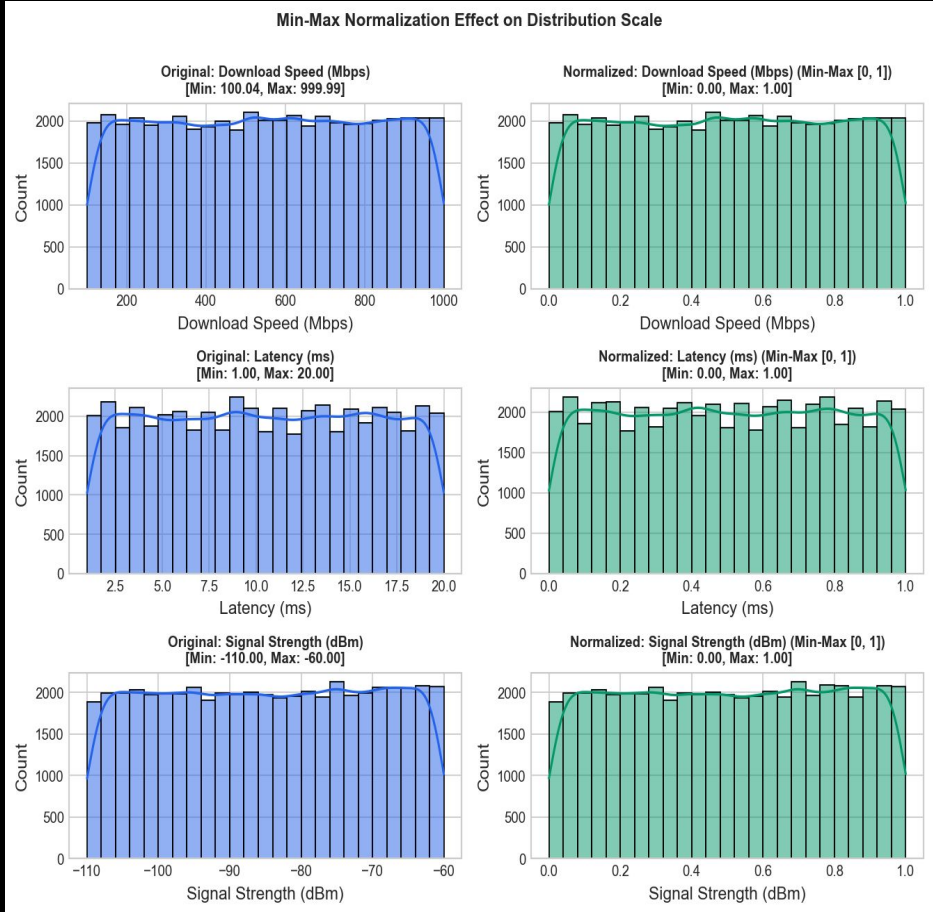
```
for col in num_cols:
    orig_missing = df[col].isnull().sum()
    mean_val = df[col].mean()
    median_val = df[col].median()
    std_val = df[col].std()
    skew_val = df[col].skew()

    df_mean_imputed[col] = df_mean_imputed[col].fillna(mean_val)
    df_median_imputed[col] = df_median_imputed[col].fillna(median_val)
```

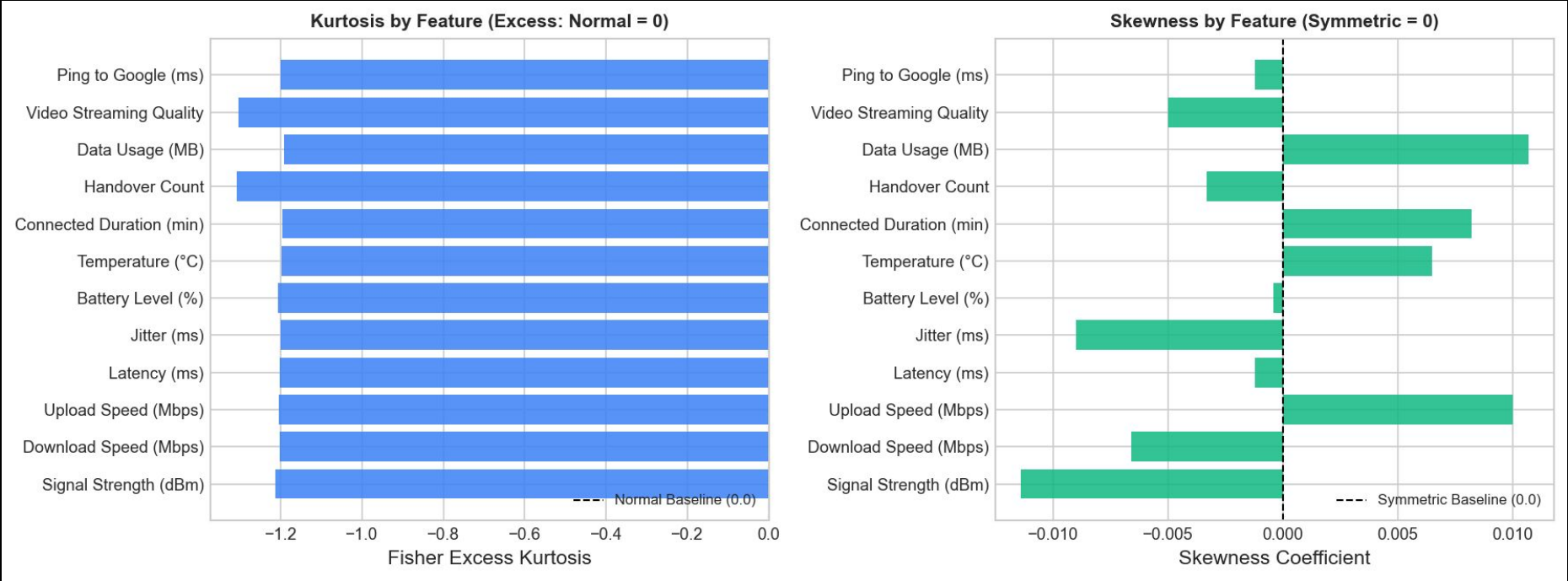
```
if abs(skew_val) > 0.75:
    recommendation = "Median (Distribution is skewed / outlier sensitive)"
else:
    recommendation = "Mean (Distribution is relatively symmetric)"
```

Step	Purpose	Outcome
df.copy()	Keeps original dataset unchanged	Original data preserved
mean()	Calculates column mean	Mean used as one imputation option
median()	Calculates column median	Median used as second option
fillna()	Replaces missing values	Missing values are filled
skew()	Checks distribution shape	Helps select Mean vs Median

Normalization and Standardization



Statistical Distribution Analysis ~Mean • Variance • Skewness • Kurtosis



THANKYOU